Exercise Sheet 01

Mistress Pell's Ravens

House rules.

- One file per exercise, in `my_work/`: `ex1_pay_maths.py`, `ex2_names.py`, …
- Start each file by typing the **Given** lines. Then write the rest yourself.
- Run it (`python my_work/ex1_pay_maths.py`) and compare with the **expected output**, character by character.
- Stuck for more than 10 minutes? Look at the cheat sheet, then skip and come back later.
- Hand in: commit and push to `lesson-01`, then open a pull request. Tell Lars which raven was the most annoying.

Exercise 1 — Pay maths

10 min · `ex1_pay_maths.py`

> **Raven from Mistress Pell (Paymaster):** *"The Grimoire failed again at the full moon. Run-time error '13', naturally. Can you redo the basic pay maths in Python? Yours in haste, P."*

GIVEN

```
annual_salary = 32500
weekly_hours = 37.5
part_time_hours = 22.5
holiday_days = 17
raise_percent = 3
years = 3
```

Calculate and print, each rounded to 2 decimal places where it makes sense:

(a) the monthly salary;

(b) the hourly rate (52 weeks a year);

(c) the salary of someone doing the same job for `part_time_hours` a week;

(d) how many full weeks and leftover days `holiday_days` are (use `//` and `%`);

(e) the salary after `years` yearly raises of `raise_percent` %, each on top of the last.

EXPECTED OUTPUT

```
Monthly salary: 2708.33
Hourly rate: 16.67
Part-time salary: 19500.0
Holiday: 2 weeks and 3 days
Salary after raises: 35513.63
```

Hint: "To the power of" is `**` in Python. If you use `^`, you'll meet it again in the Grimoire.

Exercise 2 Names and text

10 min · `ex2_names.py`

> **Raven from Orrin (Clockworks):** *"The Scriptorium's copy of the Roll is a mess. Dalvric's incantation 'cleaned' it by burning every second row. Please be gentler than Dalvric."*

GIVEN

```
raw_first = "  mAREN "
raw_last = "ashDOWN  "
raw_id = "  tm-0042 "
```

(a) Clean the names (no spaces around them, proper capitals) and print the full name.

(b) Print the initials with dots.

(c) Print the raven address: *first.last*@thornmere.realm, all lowercase.

(d) Clean the ID: no spaces, uppercase, no dash.

(e) Lady Siv's clerk is called `"Anne Marie"` `"Blackwood"`. Print her raven address, without the space.

EXPECTED OUTPUT

```
Maren Ashdown
M.A.
maren.ashdown@thornmere.realm
TM0042
annemarie.blackwood@thornmere.realm
```

Exercise 3 A pay scroll the Queen can love

10 min · `ex3_payslip.py`

> **Decree from Queen Aldara, sent at dusk:** *"The pay scrolls look 'a bit 1019'. Commas! Things lined up! Enjoy the rest day. A."*

GIVEN

```
name = "Maren Ashdown"
month = "March"
annual_salary = 32500
pension_rate = 0.05
tax_rate = 0.20
```

Basic pay is a twelfth of the salary. Guild dues are 5 % of basic pay. Crown tax is 20 % of (basic pay minus guild dues). Net pay is what's left. Print exactly this:

EXPECTED OUTPUT

```
PAY SCROLL: Maren Ashdown, March
---------------------------
Basic pay       2,708.33 cr
Guild dues        135.42 cr
Crown tax         514.58 cr
---------------------------
Net pay         2,058.33 cr

Guild dues rate: 5.0%
Annual salary: 32,500 cr
```

Hint: Each line is 27 characters: the label left-aligned in 15, the amount right-aligned in 12. Make the money text first: `basic_text = f"{basic:,.2f} cr"`, then `f"{'Basic pay':<15}{basic_text:>12}"`. And `"-" * 27` draws the line.

Exercise 4 Text that should be numbers

10 min · `ex4_conversions.py`

> **Raven from Mistress Pell:** *"The Scriptorium copies EVERYTHING out as text. Apparently '32,500.00 cr' is not a number. The Grimoire says 'Type mismatch'. I say 'please'."*

GIVEN

```
salary_text = "32,500.00 cr"
hours_text = " 37.5 "
bonus_text = "12.5%"
member_text = " Yes "
subject_number = 42
```

(a) Turn the first three into numbers and `member_text` into `True` / `False`. Print each value together with its `type()`.

(b) Build the subject ID: `TM` followed by the number padded to 4 digits.

(c) Print the bonus in crowns (salary × bonus rate).

EXPECTED OUTPUT

```
32500.0 <class 'float'>
37.5 <class 'float'>
0.125 <class 'float'>
True <class 'bool'>
TM0042
Bonus: 4,062.50 cr
```

Exercise 5 Yes or no

5 min · `ex5_true_false.py`

> **Raven from Lady Siv (Rewards):** *"Just yes or no, please. Dalvric's version once answered 'mayhap'."*

GIVEN

```
weekly_hours = 30
months_of_service = 14
rating = 4
salary = 33500
band_min = 25000
band_max = 32000
```

(a) Print four answers. **Full time**: 35 hours or more. **Bonus eligible**: at least 12 months *and* rating 3 or more. **Within band**: salary between min and max, limits included. **Needs review**: not within band.

(b) Before running it again: if `weekly_hours = 37.5` and `salary = 30000`, what will it print? Write your guess as a comment at the bottom of the file, then try it.

EXPECTED OUTPUT

```
Full time: False
Bonus eligible: True
Within band: False
Needs review: True
```

Exercise 6 LEFT, RIGHT, MID

10 min · `ex6_slicing.py`

> **Raven from Orrin:** *"In Excel I'd use LEFT/MID/RIGHT. Dalvric used four hundred lines. Please use fewer."*

GIVEN

```
realm = "Thornmere"
ref = "GUARD-1203"
```

(a) Print the Python versions of `LEFT(realm, 5)`, `RIGHT(realm, 4)` and `MID(realm, 3, 4)`.

(b) Split `ref` at the dash: print the guild, then the number as a real number.

(c) Print the number plus one, to prove it really is a number now.

EXPECTED OUTPUT

```
Thorn
mere
ornm
GUARD
1203
1204
```

Hint: Excel counts from 1, Python from 0. `ref.find("-")` tells you where the dash is. Your code should still work if the ref were `"FORGE-0007"`. Try it!

Exercise 7 The pay gap in one sentence STRETCH

15 min · `ex7_pay_gap.py`

> **Decree from Queen Aldara:** *"The Privy Council wants our pay gap 'in one sentence a queen can say out loud'. Council sits Thursday. No pressure. (Some pressure.) P.S. Biscuit's flight log is in decimal hours and nobody understands it."*

GIVEN

```
male_hourly = 20.00
female_hourly = 17.40
flight_hours = 1.75
```

The Accord's definition: **gap % = (male − female) / male × 100**, rounded to 1 decimal. Pennies per crown = female / male × 100, rounded to a whole number.

EXPECTED OUTPUT

```
Women earn 87 pennies for every crown that men earn (gap: 13.0%).
Biscuit flew for 1h 45m.
```

Hint: `int(1.75)` is `1`. The minutes come from what's left over. `{minutes:02d}` pads to two digits.

Bug hunt Pages from the Grimoire, page 1

15 min · `bughunt/payroll_bug.py`

Dalvric translated a corner of the Grimoire into Python and sailed west. Since then the raise calculation crashes, and nobody has paid crown tax for three months. **Three** bugs: one crashes, one is silently wrong, one you'll only spot letter by letter.

EXPECTED OUTPUT ONCE FIXED

```
Salary after 2 years: 31,827.00 cr
Osric Kettleby: gross 2345.50 cr, tax 469.10 cr, net 1876.40 cr
```

Commit each fix separately, explained like a detective's report. **More riddles:** five optional pages in `bughunt/riddles/`, one bug each (the last has two), getting harder as you go.